Locally recoverable codes from projective spaces.

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1. Introduction.

Locally recoverable codes provide a useful way of encoding information for applications to cloud storage. In a seminal paper, Tamo and Barg introduced a family of optimal locally recoverable codes that has captivated virtually all the attention of the Information Theory community for the past decade [Tamo-Barg 2014]. Their construction relied on so-called good polynomials, which are notoriously hard to find. We showed that there is no need to work extremely hard to come up with good polynomials: choosing certain points at random will yield optimal locally recoverable codes with probability 1.

2. Locally recoverable codes.

Definition: A linear code $\mathcal{C}$ over a finite field $\mathbb{F}_q$ is a linear subspace of $\mathbb{F}_q^n$. The parameters $[n,k,d]_q$ of the code denote its length, dimension, and minimum distance.

A code is locally recoverable with locality r when for each symbol c_i in a code word $\vec{c}=(c_1,\dots,c_n)$ there exists a recovery set $R_i \subseteq \{1,\dots,n\}\setminus\{i\}$ with $\#R_i = r$ such that c_i is a function of $\{c_j\}_{j\in R_i}$.

In particular, if the i-th coordinate of c is lost, it can be recovered by accessing $\leq r$ other symbols in $\vec{c}$.

A LRC is said to have availability t when there exist t disjoint recovery sets for each symbol c_i in $\vec{c}$.

Remark: The parameters of an LRC are constrained by a sharp Singleton-type bound for the minimum distance:

$$(SB) \qquad d \leq n-k-\left\lceil \frac{k}{r}\right\rceil+2.$$

Definition: An LRC whose parameters achieve equality in (SB) are called optimal.

Example: [Tamo-Barg 2014] Let $n\in\mathbb{N}$ and $\mathbb{F}_q$ be a finite field with $q\geq n$. Let $g(x)\in\mathbb{F}_q[x]$ be a polynomial of degree $r+1$ such that there exists a partition $A=A_1\sqcup\cdots\sqcup A_{\frac{n}{r+1}}\subseteq\mathbb{F}_q$ with $\#A=n$ and $\#A_i=r+1$ and $\#g(A_i)=1$.

That is, A is divided into $\frac{n}{r+1}$ batches and $g(x)$ is constant in each batch.

These are called <u>good polynomials</u>. Write $\vec{a} \in \mathbb{F}_q^n$ as $\vec{a} = (a_{ij})_{0 \le i \le r-1}^{0 \le j \le \frac{k}{r}-1}$, define

$$f_{\vec{a}}(x) := \sum_{i=0}^{r-1} \sum_{j=0}^{\frac{k}{r}-1} a_{ij}\, g(x)^j x^i \quad \text{and} \quad \mathcal{C} := \left\{ \left(f_{\vec{a}}(\alpha) \right)_{\alpha \in A} \;\middle|\; \vec{a} \in \mathbb{F}_q^n \right\}.$$

<u>Theorem:</u> [Tamo-Barg 2014] The code $\mathcal{C}$ is an optimal $[n,k,d]_q$ LRC with locality r.

Why is it recoverable? Because we can interpolate polynomials! Say we are missing the symbol $f_{\vec{a}}(\alpha)$ for $\alpha \in A_\ell$, then the polynomial $\partial(x) := \sum_{i=0}^{r-1} \sum_{j=0}^{\frac{k}{r}-1} a_{ij}\, g(A_i)^j x^i$ is of degree $r-1$ and we know its value at $r = \#(A_i \setminus \{\alpha\})$ points. We then know by interpolation:

$$\partial(x) = \sum_{\beta \in A_\ell \setminus \{\alpha\}} f_{\vec{a}}(\beta) \prod_{\gamma \in A_\ell \setminus \{\alpha, \beta\}} \frac{x-\gamma}{\beta-\gamma} \quad \text{and} \quad f_{\vec{a}}(x) = \partial(\alpha).$$

We have recovered $f_{\vec{a}}(\alpha)$ by only accessing the symbols in $A_\ell \setminus \{\alpha\}$.

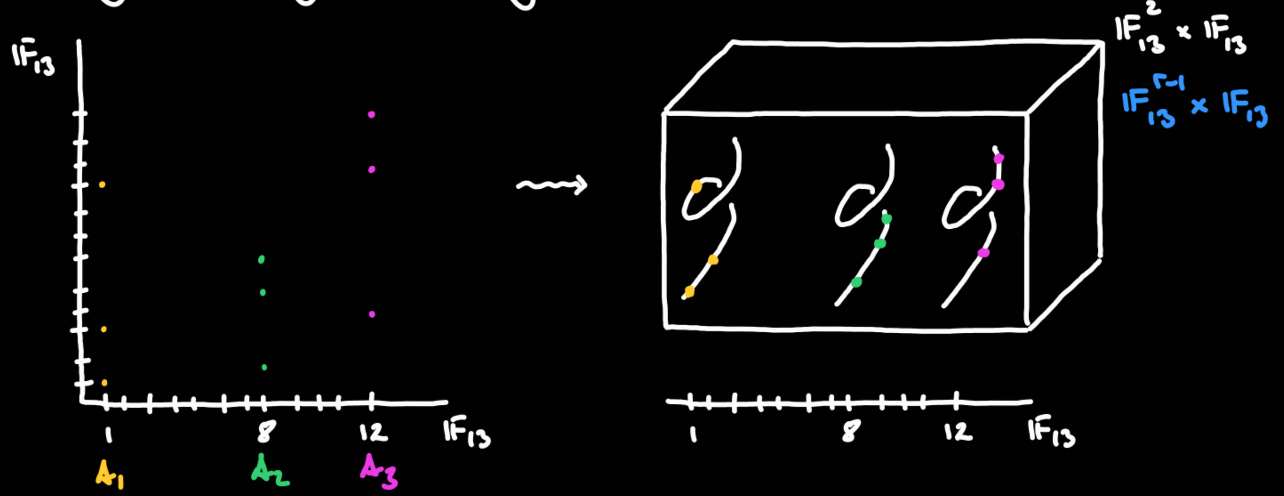
<u>Remark:</u> These codes generalize the classical Reed-Solomon codes; when $r = k$ then

$$f_{\vec{a}}(x) = \sum_{i=0}^{k-1} a_{ij} x^i \quad \text{and we recover them.}$$

Moreover, these codes can be interpreted geometrically:

$$n = 9, \quad k = 4, \quad r = 2, \quad q = 13, \quad g(x) = x^3, \quad A = \{1,3,9\} \sqcup \{2,5,6\} \sqcup \{4,10,12\}$$
$$= A_1 \sqcup A_2 \sqcup A_3.$$

Note $g(A_1) = \{1\}$, $g(A_2) = \{8\}$, $g(A_3) = \{12\}$, and we essentially have:



a geometric interpretation, when we bump up the dimension!

3. Algebro-geometric evaluation codes.

All examples and results here are from [Aguilar-Álvarez-Ardila-O.-Rodríguez-Várilly-Alvarado 2024].

<u>Definition:</u> Let X be a quasi-projective variety over $\mathbb{F}_q$, let P be a set of n points in $X(\mathbb{F}_q)$, and let V be a finite dimensional subspace of the function field of X. An <u>algebro-geometric code</u> is $\mathcal{C} := \operatorname{im} ev_P$ the image of the evaluation map:

$$ev_P : V \longrightarrow \mathbb{F}_q^n, \quad f \longmapsto (f(P_1), \dots, f(P_n)) \qquad \text{where } P = \{P_1, \dots, P_n\}.$$

__Remark:__ These codes benefit from algebro-geometric tools, for example:

$$k = \dim \mathcal{C} = \dim_{\mathbb{F}_q} \operatorname{im} ev_P = \dim_{\mathbb{F}_q} V - \dim_{\mathbb{F}_q} \ker ev_P .$$

Classically, the Riemann-Roch Theorem has been used to give bounds for their minimum distance when X is a smooth projective curve [Goppa 1977].

__Riemann-Roch Theorem:__ Let D be a divisor of X smooth projective curve, then:

$$\ell(D) \geq \deg(D) + 1 - g$$

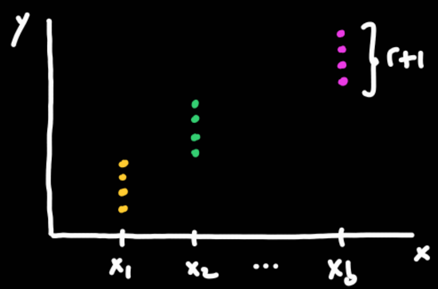
where $\ell(D)$ is the dimension of the space of rational functions associated to D, $\deg(D)$ is the degree of D, and g is the genus of X.

If you are interested in algebro-geometric codes from curves, there is a nice paper of [Munuera-Tenorio 2018], or if X being quasi-projective is too daunting.

__Example:__ Let $b, r \in \mathbb{N}$, set $n := b(r+1)$ and $k := (b-1)r$, let $\mathbb{F}_q$ be a finite field with $q \geq n$, let $X := \mathbb{A}^1_x \times \mathbb{A}^1_y$. Pick b distinct points $x_1, \dots, x_b$ in $\mathbb{A}^1_x(\mathbb{F}_q) = \mathbb{F}_q$, which we call __points on the base__, and pick n distinct points $(y_{ij})_{1 \leq i \leq b}^{1 \leq j \leq r+1}$ in $\mathbb{A}^1_y(\mathbb{F}_q) = \mathbb{F}_q$. All together form the set of points:

$$P = \{(x_i, y_{ij})\}_{1 \leq i \leq b}^{1 \leq j \leq r+1} = \{P_1, \dots, P_n\} \subseteq \mathbb{F}_q^2 = X(\mathbb{F}_q).$$

Graphically we are choosing n points in X arranged within line bundles.



Let $V := \left\{ \sum_{\ell=0}^{r-1} a_\ell(x) y^\ell \;\middle|\; a_\ell(x) \in \mathbb{F}_q[x],\ \deg a_\ell(x) \leq b-2 \right\} \subseteq \mathbb{F}_q[x]$ and define $\mathcal{C} := \operatorname{im} ev_P$ with $ev_P : V \longrightarrow \mathbb{F}_q^n$, $f(x,y) \mapsto (f(P_1), \dots, f(P_n))$. As Tamo and Barg do, we arrange the points P in batches:

$$P = A_1 \sqcup \cdots \sqcup A_b \quad \text{with } A_i = \{(x_i, y_{i1}), \dots, (x_i, y_{i\,r+1})\} \text{ for } 1 \leq i \leq b.$$

The set $\{(x_i, y) \mid y \in \mathbb{F}_q\} \subseteq X(\mathbb{F}_q)$ is called the __fiber__ above x_i for $1 \leq i \leq b$.

A __zero-fiber__ of a polynomial $f(x,y) \in V$ is a batch A_i where $f(Q) = 0$ for all $Q \in A_i$.

A __zero-fiber__ of a code word $\vec{c} = ev_P(f(x,y))$ for some $f(x,y) \in V$ is a zero-fiber of $f(x,y)$.

The points in P are in __general position__, meaning that a nonzero polynomial

$g(y) \in \mathbb{F}_q[y]$ of degree $\leq r-1$ cannot vanish along the y-coordinates of a batch of points. In fact, no two points in P share the same y-coordinate, which is a stronger condition that we use to show optimality of some of these codes.

<u>Lemma:</u> The map ev_P is injective, so $\mathcal{C}$ has dimension $k=(b-1)r$.

Why? Because if $ev_P(f(x,y)) = (0)$ then $\sum_{\ell=0}^{r-1} a_\ell(x_i) y_{ij}^\ell = f(x_i, y_{ij}) = 0$ for all $1 \leq i \leq b$ and all $1 \leq j \leq r+1$, so the polynomials $f(x_i,y) \in \mathbb{F}_q[x]$ have degree $\leq r-1$ but $r+1$ distinct zeros for all $1 \leq i \leq b$, so $f(x_i,y) \equiv 0$ for all $1 \leq i \leq b$, meaning that the polynomials $a_\ell(x) \in \mathbb{F}_q[x]$ have degree $\leq b-2$ but b distinct zeros for all $0 \leq \ell \leq r-1$, so $a_\ell(x) \equiv 0$ for all $0 \leq \ell \leq r-1$. Thus $f(x,y) \equiv 0$.

This is the kind of argument we exploit to the fullest!

<u>Lemma:</u> The code $\mathcal{C}$ has locality r.

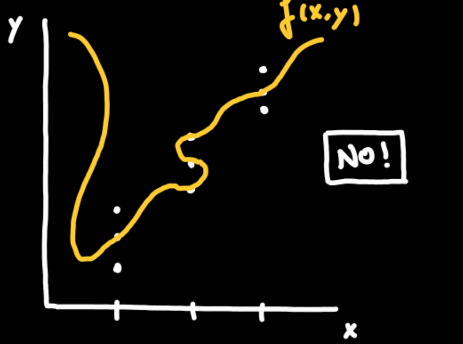
Why? Because Vandermonde matrices are invertible! Focusing on the first batch A_1, we have:

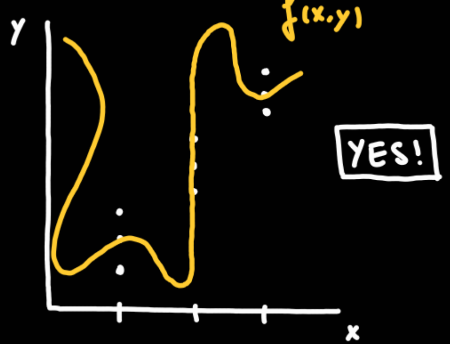
$$M \cdot a = \begin{bmatrix} 1 & y_{11} & y_{11}^2 & \cdots & y_{11}^{r-1} \\ 1 & y_{12} & y_{12}^2 & \cdots & y_{12}^{r-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & y_{1r} & y_{1r}^2 & \cdots & y_{1r}^{r-1} \\ 1 & y_{1\,r+1} & y_{1\,r+1}^2 & \cdots & y_{1\,r+1}^{r-1} \end{bmatrix} \begin{bmatrix} a_0(x_1) \\ a_1(x_1) \\ \vdots \\ a_{r-1}(x_1) \end{bmatrix} = \begin{bmatrix} f(x_1, y_{11}) \\ f(x_1, y_{12}) \\ \vdots \\ f(x_1, y_{1\,r+1}) \end{bmatrix} = F$$

giving exactly the symbols of the code word $ev_P(f(x,y))$ of the batch A_1 for a fixed polynomial $f(x,y) = \sum_{\ell=0}^{r-1} a_\ell(x) y^\ell \in V$. If we are missing the first symbol $f(x_1, y_{11})$, by deleting the first row of M and F we obtain M' and F' where it is still true that $(M') \cdot a = (F')$ but now M' is an invertible $r \times r$ Vandermonde matrix, so $a = (M')^{-1} \cdot (F')$ and we can recover $f(x_1, y) = \sum_{\ell=0}^{r-1} a_\ell(x_1) y^\ell$.

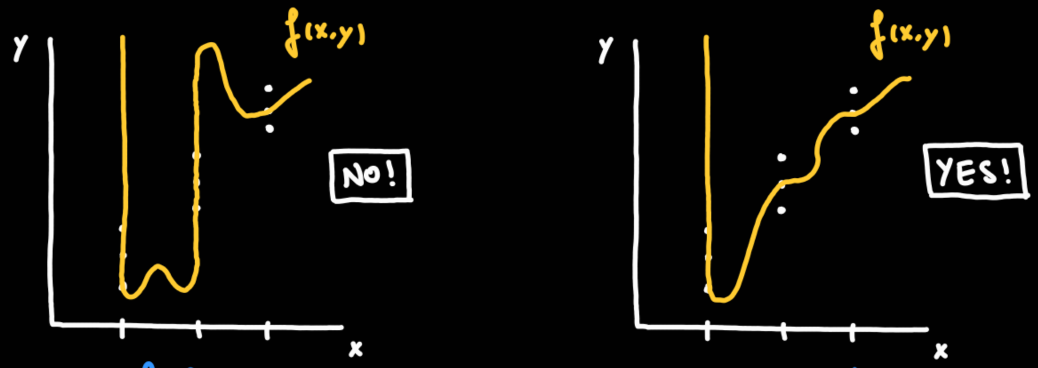
<u>Remark:</u> The minimum distance of $\mathcal{C}$ satisfies $d \leq r+3$ by the Singleton-type bound.

If A_i is a zero-fiber of $f(x,y) \in V$ then $f(x_i, y) \equiv 0$ in $\mathbb{F}_q[y]$.





Lemma: Let $b \geq 3$ and let $f(x,y) \in V$ be a nonzero polynomial. Then $f(x,y)$ has $\leq b-2$ zero-fibers.



That is, if $f(x,y)$ is nonzero, we can give a bound on how many zeros it can have. This is very useful to compute bounds for minimum distance, namely, to determine the optimality of $\mathcal{C}$.

Theorem: Let $b \geq 3$ and $r = 1,2,3$. Then $\mathcal{C}$ is an optimal LRC with parameters $n = b(r+1)$, $k = (b-1)r$, $d = r+3$, locality r, and information rate $\frac{k}{n} = \frac{b-1}{b} \cdot \frac{r}{r+1}$.

So far, we have almost exclusively used that a nonzero polynomial of degree m cannot have more than $m-1$ zeros. Here comes the algebraic geometry.

Theorem: [Lang-Weil 1954] Let $m, d, s \in \mathbb{N}$ with $d > 0$ and let $\mathbb{F}_q$ be a finite field. There exists a positive constant $A(m,d,s)$ such that for every $\mathbb{F}_q$ and every variety $X \subseteq \mathbb{P}^m$ of pure dimension s and degree d we have:

$$\# X(\mathbb{F}_q) \leq A(m,d,s) \cdot q^s.$$

Over the complex numbers, a variety X of dimension s has $\# X(\mathbb{C}) = \mathbb{C}^s$ rational points. This is not true over finite fields! The Lang-Weil estimate is saying that "the number of rational points of a variety over a finite field is roughly what you expect".

This is useful to estimate which choices of points P give optimal LRCs. The points that do not give optimal LRCs are determined by a matrix with N variables whose minors define a hypersurface $X \subseteq \mathbb{P}^{N-1}_{\mathbb{F}_q}$, so there are roughly $\# X(\mathbb{F}_q) \leq A \cdot q^{N-2}$ where A is the Lang-Weil constant. We know $\# \mathbb{P}^{N-1}(\mathbb{F}_q) = \frac{q^{N-1}-1}{q-1}$ and we need to avoid certain hyperplanes, so the points that give optimal LRCs are roughly $\# U(\mathbb{F}_q) \geq q^{N-1} + O(q^{N-2})$. Thus a randomly chosen set of points P will give nonoptimal codes with probability roughly:

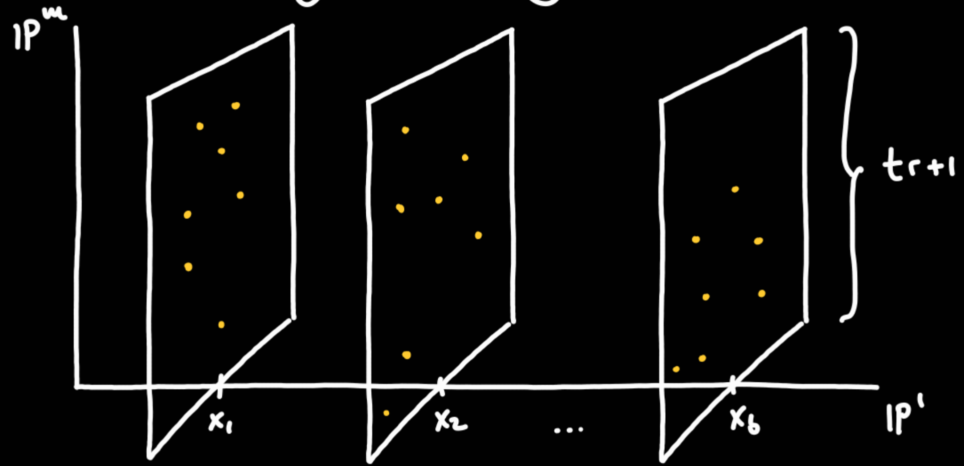
$$\frac{\# X(\mathbb{F}_q)}{\# U(\mathbb{F}_q)} \leq \frac{A\, q^{N-2}}{q^{N-1} + O(q^{N-2})} \xrightarrow[q \to \infty]{} \frac{A}{q}.$$

Theorem: Let $b \geq 3$ and $r \geq 4$. There exists $q_0 \in \mathbb{N}$ such that if $q \geq q_0$ and the points P are randomly chosen, then $\mathcal{C}$ is an optimal LRC with locality r with probability 1.

4. Codes from projective space bundles.

All results here are from [Aguilar-Álvarez-Ardila-O.-Rodríguez-Várilly-Alvarado 2024].

We can do an analogous setup in higher dimensions:



Let $m, b, t, \alpha, \beta \in \mathbb{N}$ with $b \geq \alpha+1$, let q be a prime power, set $r := \binom{\beta+m}{\beta} = \binom{\beta+m}{m}$ and $n := b(tr+1)$. Let $X := \mathbb{P}^1_x \times \mathbb{P}^m_{[y_0,\dots,y_m]}$ and $V := \Gamma(X, \mathcal{O}_X(\alpha,\beta))$ the global sections of X, which amounts to bi-homogeneous polynomials of bi-degree (α,β) with coefficients in $\mathbb{F}_q$, and $\dim_{\mathbb{F}_q} V = (\alpha+1)\binom{\beta+m}{\beta} = (\alpha+1)r$. Pick b points $x_1, \dots, x_b$ in $\mathbb{P}^1$ and for each $1 \leq i \leq b$ pick $tr+1$ distinct points in the fiber of x_i, denote them by the batch A_i. We assume the points in each batch are in <u>general position</u> within $\mathbb{P}^m$; if $g(y_0,\dots,y_m)$ is a nonzero homogeneous polynomial of degree β in $m+1$ variables, and $z_1,\dots,z_r \in A_i$, then there exists z_ℓ for $\ell \in \{1,\dots,r\}$ such that $g(z_\ell) \neq 0$. Define $ev_P : V \longrightarrow \mathbb{F}_q^n$, $f(x,y) \mapsto (f(P_1),\dots,f(P_n))$ and $\mathcal{C}_m := \operatorname{im} ev_P$.

Theorem: The code $\mathcal{C}_m$ is locally recoverable with locality r, availability t, and parameters

$$n = b(tr+1), \quad k = (\alpha+1)r, \quad d \geq (b-\alpha)((t-1)+2).$$

Theorem: The family of codes $\{\mathcal{C}_m\}_{m=1}^{\infty}$ is asymptotically good, namely:

$$\lim_{n\to\infty} \frac{d(\mathcal{C}_m)}{n(\mathcal{C}_m)} > 0 \quad \text{and} \quad \lim_{n\to\infty} \frac{k(\mathcal{C}_m)}{n(\mathcal{C}_m)} > 0.$$

Asymptotically good codes are extremely scarce in the literature, and these are the first example of high dimensional asymptotically good codes with availability.

We recover the codes we previously constructed by setting $m=1$, $t=1$, $\alpha = b-2$, $\beta = r-1$.